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Inventor(s)	Park; Sanghun

BATTERY PACK

Abstract

A battery pack includes a housing and battery modules in the housing. Each of the battery modules includes battery cells, a case accommodating the battery cells, a first fire prevention sheet on the case, and a first spacer on the case and configured to separate the first fire prevention sheet from an upper surface of the case.

Inventors:	**Park; Sanghun (Yongin-si, KR)**
Applicant:	**Samsung SDI Co., Ltd.** (Yongin-si, KR)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Patent Application No. 10-2024-0029999, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

[0002] One or more embodiments relate to a battery pack.

2. Description of the Related Art

[0003] Secondary batteries are batteries designed to be repeatedly charged and discharged and may be used as an energy source for mobile devices, electric vehicles, hybrid vehicles, electric bicycles, and uninterruptible power supplies. Depending on the type of external devices in which the secondary batteries are utilized, the secondary batteries may be utilized in the form of a single battery cell or may be utilized in the form of a module or pack in which multiple battery cells are connected to each other and bundled into a single unit.

[0004] The information disclosed in the background technology of the disclosure is only to help better understanding of the background of the disclosure, and therefore may include information that does not constitute the related art.

SUMMARY

[0005] One or more embodiments include a battery pack in which a flame may be quickly extinguished and the spread of a flame may be prevented (or at least mitigated) in response to a flame occurring in a battery module.

[0006] However, the technical objective to be solved by the present disclosure is not limited to the ones described above, and other objectives not mentioned herein may be clearly understood by those skilled in the art from the description of the disclosure described below.

[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the present disclosure.

[0008] According to one or more embodiments, a battery pack includes a housing and battery modules in the housing. Each of the battery modules includes battery cells, a case accommodating the battery cells, a first fire prevention sheet on the case, and a first spacer on the case configured to separate the first fire prevention sheet from an upper surface of the case.

[0009] The first fire prevention sheet may include two or more first fire prevention sheets, the battery modules may further include a second spacer configured to connect first fire prevention sheets to each other and to space the first fire prevention sheets apart from each other.

[0010] The first fire prevention sheets may correspond to the battery modules.

[0011] The second spacer may include a flame-retardant material, and both ends of the second spacer in a longitudinal direction may be inward relative to both ends of the first fire prevention sheet.

[0012] The first spacer may include two or more first spacers, and the first spacers may be along an edge of the case.

[0013] The first spacers may be adjacent to an edge of the case and spaced apart inward from the edge of the case.

[0014] The first spacer may include a flame-retardant material including two or more stacked layers, and an upper surface of the first spacer may be in contact with a bottom surface of the first fire prevention sheet.

[0015] The case may include a first case and a second case which is connected to the first case and the battery cells may be accommodated in the first case and the second case.

[0016] The battery pack may further include a substrate on the battery modules and electrically connected to the battery modules.

[0017] The protrusion may include two or more first upper protrusions on the upper surface of the second case, and two or more second upper protrusions on the upper surface of the second case spaced apart from the first upper protrusions.
[0018] The battery modules may each further include a first film covering a bottom surface of the first case and a second film covering an upper surface of the second case.
[0019] The second film may be spaced apart from a bottom surface of the housing.
[0020] The first fire prevention sheet may include a flame-retardant material or a fire extinguishing agent, and may cover three sides of an upper portion of the case.
[0021] The first fire prevention sheet may include a first body portion covering the upper surface of the case, and two first edge portions extending downward from two long side portions of the first body portion.
[0022] A lower end of the first edge portion may be above a center of the case in a height direction and below an upper end of the case.
[0023] The case may include a case protrusion protruding downward from a bottom surface of the case, and the housing may have a bottom surface spaced apart from the bottom surface of the case by the case protrusion.
[0024] The housing may include a housing protrusion on the bottom surface of the housing, and the housing protrusion may extend into the case protrusion or the case protrusion may extend into the housing protrusion.
[0025] The battery modules may further include a second fire prevention sheet on the bottom surface of the housing and covering all bottom surfaces of the battery modules.
[0026] The second fire prevention sheet may include a flame-retardant material or a fire extinguishing agent, and the second fire prevention sheet may wrap three sides of a lower portion of the case.
[0027] The second fire prevention sheet may include a second body portion covering a bottom surface of the case, and two second edge portions extending upward from two long side portions of the second body portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0028] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
[0029] The following drawings attached to this specification illustrate embodiments of the present disclosure, and serve to help understand the technical idea of the present disclosure along with the description of the present disclosure described later.
[0030] The disclosure is not to be construed as limited to the matters described in the drawings.
[0031] FIG. **1** illustrates a perspective view of a battery pack;
[0032] FIG. **2** illustrates a rear surface of the battery pack;
[0033] FIG. **3** illustrates a perspective view of a battery module and a substrate;
[0034] FIG. **4** illustrates a top view of the battery module and the substrate;
[0035] FIG. **5** illustrates an exploded perspective view of the battery module;
[0036] FIG. **6** illustrates a top view of the battery module;
[0037] FIG. **7** illustrates a first fire prevention sheet and a first spacer; and
[0038] FIGS. **8** and **9** illustrates cross-sections of the battery pack.
DETAILED DESCRIPTION
[0039] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In

this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” if preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0040] Some embodiments of the present disclosure and methods according thereto may be understood by referring to the detailed descriptions and drawings of the embodiments. The described embodiments may have various modifications and be implemented in different forms, and are not limited to the embodiments described herein. Additionally, some or all of the features of various embodiments of the present disclosure may be combined with each other. Each embodiment may be implemented independently or in relation to each other. The described embodiments are provided as examples to ensure that the present disclosure is thorough and complete, and are also intended to completely convey the spirit of the present disclosure to those skilled in the art to which the present disclosure pertains. This disclosure is subject to all modifications, equivalents, and substitutions within the spirit and technical scope of the present disclosure. Accordingly, processes, elements, and techniques that are not necessary to those skilled in the art for a complete understanding of the embodiments of the present disclosure may not be described.

[0041] Unless otherwise noted throughout the accompanying drawings and specification, the same reference numerals, characters, or combinations thereof indicate the same components, and thus redundant descriptions will be omitted. Additionally, in order to clearly explain the present disclosure, parts not related to the description or parts unrelated to the description have been omitted.

[0042] The relative sizes of elements, layers, and regions in the drawings may be exaggerated for clarity. The use of hatching and/or shading in the accompanying drawings generally serves to clarify boundaries between adjacent elements. Accordingly, the presence or absence of hatching or shading does not indicate a preferred form or requirement for a particular material, material properties, dimensions, proportions, commonality between figure elements, and/or any other characteristic, property, attribute, etc. of the element unless specified.

[0043] Various embodiments are described herein with reference to cross-sectional examples that are schematic illustrations of embodiments and/or intermediate structures. The appearance of the drawing may therefore vary, for example as a result of manufacturing techniques and/or tolerances. In addition, the specific structural or functional description disclosed in this specification is merely an example for explaining embodiments according to the concept of the present disclosure. Accordingly, the embodiments disclosed in this specification should not be construed as being limited to the shape of the illustrated areas and include, for example, variations in shape due to manufacturing.

[0044] The areas illustrated in the drawings are schematic in nature and their shapes are not intended to be limiting or illustrative of the actual shape of the device areas. Additionally, as those skilled in the art will recognize, the described embodiments may be modified in various ways without departing from the spirit or scope of the present disclosure.

[0045] Numerous specific details are set forth in the specification to provide a thorough understanding of the various embodiments. However, various embodiments may be practiced without or including one or more of these specific details. In other instances, well-known structures and devices are illustrated in block diagram form to avoid unnecessarily obscuring the various embodiments.

[0046] For ease of explanation herein, to describe the relationship of one element or feature to another as illustrated in the drawings, spatially relative terms such as “below”, “above”, “lower”, “upper”, etc. may be used. Spatially relative terms are intended to include various orientations of the device in use or operation in addition to those illustrated in the drawings. For example, if a device in the drawings is turned over, other elements or features described as being “below” or “under” will be facing “above” the other elements or features. Accordingly, as illustrative terms, “below” and “under” may include both up and down directions. A device may be oriented in different directions (e.g., rotated 90 degrees or in another direction) and the spatially relative descriptions used herein should be interpreted accordingly. Similarly, if the first part is described as being disposed “above” the second part, this means that the first part is disposed above or below the second part.

[0047] In addition, the expression “viewed from a plane” refers to a view of an object from above, and the expression “in a schematic cross-sectional view” refers to a schematic cross-section taken by cutting the object vertically. The term “viewed from the side” indicates that a first object may be above, below or to the side of a second object and vice versa. Additionally, the term may include layers, lamination, surface, extension, covering, or partially covering, or any other suitable term that would be understood and understood by those skilled in the art. The expression “does not overlap” may include meanings such as “being distant from” or “spaced apart from” and any other suitable equivalents recognized and understood by those skilled in the art. The terms “side” and “surface” may indicate that a first object may directly or indirectly face a second object. If there is a third object between the first object and the second object, the first object and the second object face each other, but may be understood as indirectly opposing each other.

[0048] If an element, layer, region or component is referred to as being “formed”, “connected” or “coupled” on or to another element, the element, layer, region or component may be formed directly on the element, layer, region or component, or may be formed on another element, layer, region or component, or indirectly formed on, connected to, or coupled to other components. The above expression may also collectively refer to direct or indirect combinations or connections of elements, layers, regions or components and integral or non-integral combinations or connections so that one or more elements, layers, regions or components can exist. For example, if an element, layer, region or component is referred to as being “electrically connected” or “electrically coupled” to another element, layer, region or component, this may indicate that the element, layer, region or component is directly electrically connected or coupled thereto, or other elements, layers, regions or components may be present. However, “directly connected” or “directly coupled” indicates that one component is directly connected or coupled to another component without an intermediate component, or is on another component. In addition, in this specification, if a portion of a layer, film, region, plate, etc. is formed on another part, the formation direction thereof is not limited to an upper direction, and includes the portion being formed on the side or on the bottom. Conversely, if a portion of a layer, film, region, plate, etc. is formed “under” another part, not only is that portion “immediately below” the other part, but also there may be another part between the part and the other part. Other expressions that describe relationships between components, such as “between,” “immediately between,” “adjacent to,” and “immediately adjacent to,” may be interpreted similarly. Additionally, if an element or layer is referred to as being “between” two elements or layers, the element or the layer may be the only element between the two elements or layers, or there may be other elements therebetween.

[0049] For the purposes of this specification, expressions such as “at least one or more” or “any one” do not limit the order of individual elements. For example, “at least one of X, Y and Z”, “at least one of X, Y or Z”, “at least one selected from the group consisting of X, Y, and Z” may include any combination of two or more of X, Y and Z. Similarly, expressions such as “at least one of A and B” and “at least one of A or B” may include A, B or A and B. As used herein, the term “or” generally refers to “and/or” and “and/or” includes any combination of one or more related list items. For example, an expression such as “A and/or B” may include A, B, or A and B.

[0050] Although the terms “first”, “second”, “third”, etc. may be used herein to describe various elements, components, regions, layers and/or sections, such elements, components, regions, layers and/or sections are not limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Accordingly, a first element, component, region, layer or section described below may be referred to as a second element, component, region, layer or section without departing from the spirit and scope of the present disclosure. Describing an element as a “first” element may not require or imply the presence of a second or another element. Terms such as “first,” “second,” etc. may be used herein to distinguish different categories or sets of elements. For clarity, terms such as “first,” “second,” etc. may refer to “first category (or first set),” “second category (or second set),” etc., respectively.

[0051] The terms used in this application are only used to describe certain embodiments and are not intended to limit the present disclosure. As used herein, singular terms are intended to include plural terms and plural terms are also intended to include the singular, unless the context clearly dictates otherwise. The terms “include,” “comprise,” and “have” if used herein are meant to designate the presence of specified features, integers, or steps. These expressions do not exclude the presence or addition of one or more other functions, steps, operations, components and/or groups thereof.

[0052] If one or more embodiments may be implemented differently, certain process sequences may be performed differently than the described order. For example, two processes described in succession may be performed substantially simultaneously or may be performed in an order opposite to that described.

[0053] As used herein, the term “substantially,” “about,” “approximately,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent deviations in measured or calculated values that would be recognized by those of ordinary skill in the art. “About” or “approximately,” as used herein, is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, “about” may mean within one or more standard deviations, or within ±30%, 20%, 10%, 5% of the stated value. Further, the use of “may” when describing embodiments of the present disclosure refers to “one or more embodiments of the present disclosure.”

[0054] Unless defined differently, all terms used in the description including technical and scientific terms have the same meaning as generally understood by those skilled in the art to which the present disclosure pertains. Terms as defined in a commonly used dictionary should be construed as having the same meaning as in an associated technical context and/or the context of the present specification, and unless defined apparently herein, the terms are not ideally or excessively construed as having formal meaning.

[0055] FIG. **1** illustrates a perspective view of a battery pack **10**, FIG. **2** illustrates a rear surface of the battery pack **10**, FIG. **3** illustrates a perspective view of a battery module **300** and a substrate **400**, FIG. **4** illustrates the battery module **300** and the substrate **400**, FIG. **5** illustrates an exploded perspective view of the battery module **300**, FIG. **6** illustrates a top view of the battery module **300**, FIG. **7** illustrates a first fire prevention sheet **360** and a first spacer **370**, and FIGS. **8** and **9** illustrate a cross-section of the battery pack **10**. For example, FIGS. **8** and **9** illustrate cross-sections parallel to a YZ plane, and for convenience of description, the battery module **300** on the left in FIG. **8** represents a state without a battery cell C, and the battery module **300** on the right of FIG. **8** illustrates a state including a battery cell C. FIG. **9** illustrates a cross-section of the battery pack **10** in which the battery modules **300** are arranged in a different direction from that of FIG. **8**.

[0056] The battery pack **10** may be applied to large applications such as an energy storage system (ESS) or electric vehicles, or small to medium-sized applications such as power tools or smartphones. For example, the battery pack **10** may be coupled to other devices. In one or more embodiments, the battery pack **10** may supply power to an external device through a first adapter

A**1**. The battery pack **10** may be physically connected to an external device through a second adapter A**2**. The external device may be a power tool such as a chain saw, drill, lawn mower, etc.

[0057] The battery pack **10** may be configured to be worn or carried by a user. In one or more embodiments, the battery pack **10** may be a backpack that a user can wear. A connection member such as a string, buckle, or belt may be attached to the battery pack **10** (e.g., the back surface of the battery pack **10**). A user may carry the battery pack **10** by connecting the connection member to the body, such as the shoulder, waist, or the like, and connect the battery pack **10** to an external device, such as a power tool, to supply power to the external device. The battery pack **10** may include a handle H so that the user can carry the battery pack **10**. In FIG. **1**, the handle H on an upper portion of the housing **100** is illustrated, but one or more handles H may be on a front surface of the housing **100** (e.g., a side facing a Z-axis direction of FIG. **1**), a rear surface, a side surface, and/or a bottom surface of the housing **100**.

[0058] The battery pack **10** may include the housing **100** and a plurality of battery modules **300** included in the housing **100**, and each battery module **300** may include a plurality of battery cells C, a case **310** including the plurality of battery cells C, a first fire prevention sheet **360** on the case **310**, and a first spacer **370** which is on the case **310** and separates the first fire prevention sheet **360** from an upper surface of the case **310**.

[0059] The battery pack **10** may include the housing **100**, a cover **200**, the battery module **300**, and the substrate **400**.

[0060] The housing **100** may hold and/or support other components of the battery pack **10** (e.g., the cover **200**, the battery module **300**, and the substrate **400**). The housing **100** may be at the rear of the battery pack **10** and may be hollow. The battery module **300** may be inside the housing **100**. The housing **100** may be hollow and have an open upper surface. The open upper surface of the housing **100** may be covered by the cover **200**. However, the shape of the housing **100** is not limited thereto and may vary depending on the shape of the battery pack **10** and/or the size and capacity of the battery module **300**, etc.

[0061] The housing **100** may include a housing protrusion **110** (see FIGS. **8** and **9**). One or more housing protrusions **110** may be formed on the bottom surface of the housing **100** (for example, a surface facing the battery module **300**) and may be connected to the battery module **300**. In one or more embodiments, as illustrated in FIG. **8**, a case protrusion **3111** of the case **310** (e.g., a first case **311**) of the battery module **300** may be inserted into the housing protrusion **110**, and the battery module **300** may be fixed to the housing **100**. In one or more embodiments, the housing protrusion **110** may have a hollow cylindrical shape, and the case protrusion **3111** may be inserted into the housing protrusion **110**. The housing protrusion **110** may separate the battery module **300** (e.g., the first case **311**) from the bottom surface of the housing **100**. Accordingly, if the battery cell C ignites, gas generated from the battery cell C may be discharged easily. In one or more embodiments, as illustrated in FIG. **8**, the bottom surface of the housing **100** and a bottom surface of the first case **311** (or a bottom surface of the first film **340**) may be spaced apart from each other by a gap having a distance L**4**, and thus, gas generated in a lower portion of the battery cell C (e.g., a bottom electrode) may be easily discharged via the gap having the distance L**4**. Thus, an excessive increase in the internal pressure of the battery module **300** due to gas generated from a fire occurring in the battery cell C may be prevented or at least mitigated. In FIG. **8**, two housing protrusions **110** respectively corresponding to two battery modules **300**, but the number of housing protrusions **110** is not limited to two. In one or more embodiments, the housing **100** may include a plurality of housing protrusions **110** corresponding in number to a plurality of battery modules **300**, and the number of housing protrusions **110** may vary according to the size, arrangement, and shape of the battery modules **300**.

[0062] In one or more embodiments, the housing protrusion **110** may not be directly fastened to the battery module **300** but may only separate the battery module **300** apart from the housing **100**. In the embodiment illustrated in FIG. **9**, the housing protrusion **110** may have a pin or thin plate shape protruding from the bottom surface of the housing **100**. An upper end of the housing protrusion **110** may contact the bottom surface of the case **310** so that the bottom surface of the case **310** is spaced apart from the bottom surface of the housing **100**.

[0063] The housing **100** may include a first partition wall **120**. The first partition wall **120** may protrude from the bottom surface of the housing **100** in a height direction (e.g., the Z-axis direction in FIG. **9**) and separate adjacent battery modules **300** apart from each other in a width direction of the housing **100** (e.g., a Y-axis direction in FIG. **9**). In the embodiment illustrated in FIG. **9**, the first partition wall **120** may be at a center (or substantially a center) in the width direction of the housing **100**. The first partition wall **120** may divide the internal space of the housing **100** into spaces where each battery module **300** is accommodated, and the first partition wall **120** may be spaced apart from an inner side surface of the battery module **300** to allow gas generated from the battery cell C to be easily discharged.

[0064] The housing **100** may include a second partition wall **130**. The second partition wall **130** may divide an internal space of the housing **100** including the battery module **300** and may be connected to the cover **200**. In the embodiment illustrated in FIG. **9**, the battery module **300** may be in partitioned spaces between the first partition wall **120** and the second partition wall **130**. The second partition wall **130** may be spaced apart from an outer side surface of the battery module **300**. Accordingly, gas generated from the battery cell C may be easily discharged.

[0065] The cover **200** may be connected to the housing **100**. In the embodiment illustrated in FIG. **1**, the cover **200** may cover the open upper surface of the housing **100** and the battery module **300** and the substrate **400** may be accommodated in the housing **100**. The cover **200** may be hollow and have an open bottom surface corresponding to an upper surface of the housing **100**. The cover **200** may be attachable/detachable to/from the housing **100**.

[0066] One or more battery modules **300** may be included in the battery pack **10**. The battery module **300** may include a plurality of battery cells C and may supply power to an external device through the first adapter A**1**. In the embodiment illustrated in FIG. **2**, two battery modules **300** may be included in the battery pack **10**. The two battery modules **300** may be spaced apart from each other and may not contact each other. However, the number of battery modules **300** is not limited to two, and any other suitable number of battery modules **300** may be included depending, for example, on the specifications and application of the battery pack **10**. The battery cells C included in the battery module **300** may be at least one of a cylindrical shape, a prismatic shape, and/or a pouch shape. Hereinafter, for convenience of description, the description will focus on an embodiment in which the battery cell C is cylindrical.

[0067] The battery module **300** may include the case **310**, a bus bar **320**, a cell tab **330**, the first film **340**, the second film **350**, the first fire prevention sheet **360**, the first spacer **370**, the second fire prevention sheet **380**, and a second spacer **390**.

[0068] The case **310** is configured to hold and support the battery cell C and allow the battery module **300** to be supported in the housing **100**. In one or more embodiments, the case **310** may be hollow and have a substantially rectangular parallelepiped shape. A lower portion of the case **310** may face the bottom surface of the housing **100**, and the lower portion and an upper portion of the case **310** may be wrapped by the first film **340** and the second film **350**, respectively. Additionally, the upper portion and the lower portion of the case **310** may be wrapped by the first fire prevention sheet **360** and the second fire prevention sheet **380**, respectively. The substrate **400** may be on the case **310**.

[0069] The case **310** may be arranged such that electrodes of the battery cells C face the bottom surface of the housing **100** and the substrate **400**. In the embodiment illustrated in FIG. **8**, the case **310** may be arranged such that an upper end and a lower end of the battery cells C face a bottom surface of the substrate **400** and the bottom surface of the housing **100**, respectively. Accordingly, in response to one of the battery cells C igniting, gas generated from the electrodes of the battery cells C may be easily discharged through the gap having the distance L**4** between the bottom surface of the housing **100** and the first film **340** and gap having the distance L**3** between the substrate **400** and the second film **350**, thereby preventing (or at least mitigating) an excessive pressure increase in the battery module **300** and additional damage to the battery module **300** and the battery pack **10**.

[0070] The case **310** may be arranged such that the electrodes of the battery cells C face a side surface of the housing **100**. In the embodiment illustrated in FIG. **9**, the case **310** may be arranged such that the upper end and the lower end thereof corresponding to the electrodes of the battery cells C face the inner side surface of the housing **100**. An inner side surface of the case **310** may be spaced apart from the first partition wall **120**, and an outer side surface of the case **310** may be spaced apart from the second partition wall **130**. Accordingly, in response to one of the battery cells C igniting, gas generated from the electrodes of the battery cells C may be easily discharged through a distance (e.g., a gap) between a side surface of the case **310** and the first partition wall **120** and the second partition wall **130**, thereby preventing (or at least mitigating) spreading of a flame and thereby reducing additional damage to the battery module **300** and battery pack **10**.

[0071] The case **310** may include the first case **311** and the second case **312**.

[0072] The first case **311** may be in a lower portion of the case **310** and may be a portion directly connected to or in contact with the bottom surface of the housing **100**. For example, the first case **311** and the second case **312** may include a plurality of battery cells C therein. The first film **340** may be attached to the bottom surface of the first case **311** for insulation.

[0073] The first case **311** may include the lower protrusion **3111**. In the embodiment illustrated in FIG. **8**, one or more lower protrusions **3111** may be on the bottom surface of the first case **311** and may protrude toward the bottom surface of the housing **100**. At least a portion of the lower protrusion **3111** may extend into the housing protrusion **110** of the housing **100**, or the housing protrusion **110** may extend into the lower protrusion **3111**, thereby fixing (e.g., coupling) the first case **311** to the housing **100**. The first case **311** may be spaced apart from the bottom surface of the housing **100** by a gap having the distance L**4** due to the lower protrusion **3111** and the housing protrusion **110**, and thus, gas generated from the bottom surface of the battery cell C may be easily discharged.

[0074] The second case **312** may be in the upper portion of the case **310** and may include a plurality of battery cells C inside the second case **312** and the first case **311**. The second case **312** may be attachable/detachable to/from the first case **311** (e.g., the second case **312** may be detachably coupled to the first case **311**). The second film **350** may be attached to an upper surface of the second case **312** for insulation.

[0075] The second case **312** may include a plurality of protrusions. In one or more embodiments, the second case **312** may include a plurality of first upper protrusions **3121** and a plurality of second upper protrusions **3122** on an upper surface thereof. The plurality of protrusions may separate the substrate **400** apart from the upper surface of the second case **312**. Additionally, the second case **312** may include an insertion hole **3123**, a first fastening hole **3124**, and a second fastening hole **3125**.

[0076] One or more first upper protrusions **3121** may be on the second case **312**. In the embodiment illustrated in FIGS. **4** and **6**, a plurality of first upper protrusions **3121** may be on the upper surface of the second case **312**. The first upper protrusion **3121** may pass through the second film **350** and extend into a first sheet hole **361** of the first fire prevention sheet **360**. The first upper protrusion **3121** may extend into the first sheet hole **361** to fix the second film **350** and the first fire prevention sheet **360** on the second case **312**. In one or more embodiments, the first upper protrusion **3121** may include a cylinder or a truncated cone shape tapered in an upward direction. One or more ribs may be on an outer peripheral surface of the first upper protrusion **3121**, and the first sheet hole **361** may include the ribs and have a shape corresponding to the first upper protrusion **3121**. The ribs may extend into the first sheet hole **361** such that the first fire prevention sheet **360** may be firmly supported in the second case **312**.

[0077] The plurality of first upper protrusions **3121** may be along a long side portion of the second case **312**. In the embodiment illustrated in FIG. **6**, three first upper protrusions **3121** may be on each long side portion of the second case **312** (i.e., a total of six protrusions on the second case **312**). However, the number of first upper protrusions **3121** is not limited to six, and the number of first upper protrusions **3121** may vary depending, for example, on the size and shape of the second case **312**, the second film **350**, and the first fire prevention sheet **360**. At least some of the plurality of first upper protrusions **3121** may be connected to the substrate **400**. In the embodiment illustrated in FIG. **3**, at least some of the plurality of first upper protrusions **3121** on an inner long side portion among a pair of long side portions of the second case **312** and at least some of the plurality of first upper protrusions **3121** on an outer long side portion among the pair of long side portions of the second case **312** may extend into the substrate **400**. Thus, the substrate **400** and the second case **312** may be firmly connected to each other. Due to the first upper protrusion **3121**, the substrate **400** is separated or spaced apart from the first fire prevention sheet **360**, thereby minimizing (or at least reducing) heat transferred to the substrate **400** in response to a fire occurring in the battery module **300**.
[0078] One or more second upper protrusions **3122** may be on the second case **312**. In the embodiment illustrated in FIGS. **4** and **6**, a plurality of second upper protrusions **3122** may be on the upper surface of the second case **312**. The second upper protrusions **3122** may be spaced apart from the first upper protrusions **3121**. The second upper protrusion **3122** may extend into the substrate **400** after passing through the second film **350** and the first fire prevention sheet **360**. The second upper protrusion **3122** may fix the second film **350**, the first fire prevention sheet **360**, and the substrate **400** on the second case **312**. In one or more embodiments, the second upper protrusion **3122** may have a cylindrical shape. The plurality of second upper protrusions **3122** may in a center portion (or substantially a center portion) of the second case **312**. In the embodiment illustrated in FIG. **6**, two second upper protrusions **3122** may be on an inner portion of the second case **312** in a width direction (e.g., the Y-axis direction in FIG. **6**) with respect to the first upper protrusion **3121**. However, the number of second upper protrusions **3122** is not limited to two, and the number of first upper protrusions **3121** may vary depending, for example, on the size and shape of the second case **312**, the second film **350**, the first fire prevention sheet **360**, and the substrate **400**.
[0079] An upper portion of at least some of the plurality of first upper protrusions **3121** and an upper portion of at least some of the plurality of second upper protrusions **3122** may extend into the substrate **400** to support the substrate **400** and separate or space the substrate **400** apart from the upper surface of the first fire prevention sheet **360**.
[0080] Each of the insertion holes **3123** is a portion in which the battery cell C is accommodated, and the insertion hole **3123** may have a circular shape corresponding to the shape of the battery cell C. The insertion hole **3123** of the second case **312** may correspond to an insertion hole of the first case **311**, and may define, together with the insertion hole of the first case **311**, a space in which one of the battery cells C is accommodated. The number of insertion holes **3123** may be equal to the number of battery cells C.
[0081] The first fastening hole **3124** and the second fastening hole **3125** may each be connected to the bus bar **320**. In the embodiment illustrated in FIG. **6**, the first fastening hole **3124** and the second fastening hole **3125** may be connected to a first bus bar **321** and a second bus bar **322**, respectively, through a fastening member F. One or more first fastening holes **3124** and one or more second fastening holes **3125** may be in the upper surface of the second case **312**, and may be portions where the bus bar **320** and the cell tab **330** are connected to each other through the fastening member F. In the embodiment illustrated in FIG. **6**, one first fastening hole **3124** may be on each of a pair of long side portions of the second case **312**, and a plurality of second fastening holes **3125** (e.g., three) may be on one of the long side portions of the second case **312**. One first fastening hole **3124** may be connected to the first bus bar **321**, and the other first fastening hole

3124 and the remaining plurality of second fastening holes **3125** may be connected to the second bus bar **322**.

[0082] The bus bar **320** may include a conductive material such as metal and may be electrically connected to the cell tab **330**. The bus bar **320** may include a first bus bar **321** and a second bus bar **322**. The first bus bar **321** and the second bus bar **322** may each be electrically connected to different cell tabs **330**. The bus bar **320** may be connected to the substrate **400**, etc., and may electrically connect the plurality of battery cells C and the substrate **400** to each other.

[0083] The cell tabs **330** may electrically connect the plurality of battery cells C to each other. In one or more embodiments, the cell tabs **330** may be in an upper portion and a lower portion of the battery module **300**, respectively, and may be connected to the electrodes of the battery cells C exposed through the bottom surface and the upper surface of the first case **311** and the second case **312**. A plurality of cell tabs **330** may be included in each battery module **300**. In the embodiment illustrated in FIG. **5**, the cell tabs **330** may include a plurality of first cell tabs **331** on a bottom surface of the first case **311** and a plurality of second cell tabs **332** on the upper surface of the second case **312**. Each first cell tab **331** may be connected to the lower portion of the plurality of battery cells C and electrically connected to the plurality of battery cells C, and may be connected to the bus bar **320**. Each second cell tab **332** may be connected to the upper portion of the plurality of battery cells C, electrically connected to the plurality of battery cells C, and connected to the bus bar **320**.

[0084] The first film **340** and the second film **350** are components configured to insulate the battery module **300** and may be in a lower portion and an upper portion of the battery module **300**, respectively. In the embodiment illustrated in FIG. **5**, the first film **340** may be under the first case **311**, and the second film **350** may be on (e.g., above) the second case **312**. The first film **340** may cover the bottom surface of the first case **311** and the second film **350** may cover the upper surface of the second case **312**. The first film **340** and the second film **350** may cover the first cell tab **331** and the second cell tab **332**, respectively, so that the electrodes of the battery cells C do not contact the battery module **300** or other components of the battery pack **10**. In one or more embodiments, the first film **340** and the second film **350** may have a thin flat plate shape. In one or more embodiments, the first film **340** and the second film **350** may each have a surface coated with adhesive (e.g., acrylic adhesive), each surface facing the first case **311** and the second case **312**, respectively, and the opposite surfaces thereof may include an insulating tape including an insulating material such as propylene, polyester, or polyimide. In one or more embodiments, the first film **340** and the second film **350** may have a relatively smaller thickness than at least some other components of the battery module **300** (e.g., the case **310**, the bus bar **320**, the cell tab **330**, the first fire prevention sheet **360**, the first spacer **370**, the second fire prevention sheet **380**, and the second spacer **390**), and the thickness of the first film **340** and the second film **350** may be negligible compared to the thicknesses of these other components of the battery module **300**. In the embodiment illustrated in FIG. **8**, the distance L**3** between an upper surface of the second film **350** and the first fire prevention sheet **360** may be substantially equal to a distance between the upper surface of the second case **312** and the first fire prevention sheet **360**, and these two distances may be used in the present specification with substantially the same meaning. In the embodiment illustrated in FIG. **8**, the distance L**4** between the bottom surface of the first film **340** and the second fire prevention sheet **380** may be substantially equal to a distance between the bottom surface of the first case **311** and the second fire prevention sheet **380**, and these two distances may be used in the present specification with substantially the same meaning.

[0085] The first fire prevention sheet **360** may be in the upper portion of the battery module **300** and may be configured to prevent (or at least mitigate) the spread of flames generated by the battery module **300**. In one or more embodiments, the first fire prevention sheet **360** may include a fire extinguishing agent. In response to a flame occurring in the battery cell C, a portion of the first fire prevention sheet **360** may be configured to be torn or ruptured due to heat and pressure and the fire extinguishing agent may be sprayed into the battery cell C. The first fire prevention sheet **360** may include a flame-retardant material and may be configured to prevent (or at least mitigate) flames from spreading to other components of the battery pack **10**, such as the substrate **400**. In one or more embodiments, the first fire prevention sheet **360** may include ceramic paper, mica, or aerogel. The first fire prevention sheet **360** may be on the second film **350** and cover the battery module **300**. The first fire prevention sheet **360** may cover three upper sides of the case **310**. In one or more embodiments, the first fire prevention sheet **360** may wrap the upper surface of the case **310** (e.g., the upper surface of the second case **312**) and two of a plurality of side surfaces of the case **310**, which are connected to the upper surface of the case **310** (e.g., two side surfaces corresponding to the long side portions of the second case **312**).

[0086] The battery pack **10** may include a plurality of first fire prevention sheets **360**. In one or more embodiments, the first fire prevention sheet **360** may include the same number of first fire prevention sheets **360** as the number of the plurality of battery modules **300**, and the plurality of first fire prevention sheets **360** may respectively correspond to the plurality of battery modules **300** (e.g., one first fire prevention sheet **360** for each battery module **300**). Therefore, by reducing the size and weight of the first fire prevention sheet **360** included in each battery module **300**, sagging and damage to the first fire prevention sheet **360** may be prevented (or at least mitigated against). Different first fire prevention sheets **360** respectively corresponding to the battery modules **300** may be connected through the second spacer **390**.

[0087] The first fire prevention sheet **360** may be spaced apart from the case **310** and/or the second film **350**. In the embodiment illustrated in FIG. **8**, one or more first spacers **370** may be formed on the second film **350**, and the upper surface of the second film **350** and the first fire prevention sheet **360** may be spaced apart from each other by the first spacer **370** by a gap having the distance L**3**. Therefore, gas generated from the battery cell C may be easily discharged. By separating the first fire prevention sheet **360** and the second film **350** apart from each other, the fire extinguishing agent included in the first fire prevention sheet **360** may be easily sprayed over a relatively wide area.

[0088] The first fire prevention sheet **360** may include a first body portion **3601** and a first edge portion **3602**. The first body portion **3601** may be supported by the first spacer **370** and may have a larger area than the case **310**. The first body portion **3601** may be attached to the upper surface of the second case **312**, and the first body portion **3601** may cover the plurality of second cell tabs **332**. The first edge portion **3602** may extend downward from an edge or edges of the first body portion **3601**. In one or more embodiments, the first edge portion **3602** may correspond to one or both of the long side portions of the first body portion **3601**. As illustrated in FIG. **3**, the first edge portion **3602** may extend downward from a pair of long side portions of the first body portion **3601** and surround at least a portion of the upper portion of the case **310**. The first edge portion **3602** may prevent (or at least mitigate) flames from spreading in a lateral direction (e.g., the Y-axis direction in FIG. **3**).

[0089] In the embodiment illustrated in FIG. **3**, the first edge portion **3602** may not be on either of the short side portions of the first body portion **3601**. The first edge portion **3602** may be formed on the pair of long side portions of the first body portion **3601** and may not be formed on a pair of short side portions thereof. As illustrated in FIG. **3**, in a longitudinal direction of the battery module **300** (e.g., X-axis direction in FIG. **3**), a front surface and a rear surface of the battery module **300** or the case **310** may not be covered by the first fire prevention sheet **360**. In one or more embodiments, the first fire prevention sheet **360** may cover three sides of the battery module **300** or the case **310** (e.g., the second case **312**). In one or more embodiments, short side portions of the case **310** may not be covered by the first fire prevention sheet **360**. Therefore, the first fire prevention sheet **360** may prevent (or at least mitigate) the spread of flames generated from the long side portion of the battery module **300** and allow a small amount of gas to be easily discharged from the short side portion of the battery module **300**. Additionally, the flow of gas may be controlled such that the gas generated in the battery module **300** is discharged to the short side portion of the battery module **300**.

[0090] A lower end of the first edge portion **3602** may be above a center of the case **310** in a height direction and below an upper end of the case **310**. In one or more embodiments, a lower end of the first edge portion **3602** may be above a connection portion between the first case **311** and the second case **312**, and may be below an upper end of the second case **312**. If the first edge portion **3602** extends to the connection portion between the first case **311** and the second case **312**, the area covered by the first edge portion **3602** becomes too wide, causing an excessively high internal pressure of the battery module **300** in response to a flame occurring in the battery module **300**. In an embodiment in which the first edge portion **3602** is above the upper end of the second case **312**, the effect of preventing (or at least mitigating) flame spread from the long side portion of the battery module **300** may decrease. An inner side surface of the first edge portion **3602** may be spaced apart from the side surface of the case **310**.

[0091] The first fire prevention sheet **360** may include a first sheet hole **361**, a second sheet hole **362**, and a third sheet hole **363**.

[0092] As illustrated in FIG. **7**, a plurality of first sheet holes **361**, second sheet holes **362**, and third sheet holes **363** may each be in the first fire prevention sheet **360**, and may be portions connected to the case **310**, the bus bar **320**, and the substrate **400**. In one or more embodiments, the plurality of first sheet holes **361** may be in the first fire prevention sheet **360** to correspond to the first upper protrusions **3121** of the second case **312**, and the plurality of second sheet holes **362** may be in the first fire prevention sheet **360** to correspond to the second upper protrusions **3122** of the second case **312**. The third sheet holes **363** are configured to fasten the bus bar **320** to the second case **312**. In one or more embodiments, the third sheet holes **363** may correspond to the first fastening holes **3124** to expose the first fastening holes **3124**.

[0093] One or more first spacers **370** may be included in the battery module **300** and separate the first fire prevention sheet **360** and the second film **350** apart from each other. The plurality of first spacers **370** may be on the upper surface of the second film **350**, and an upper surface of the first spacers **370** may be in contact with a bottom surface of the first fire prevention sheet **360**. The first spacer(s) **370** may have a lower height than both the first upper protrusion **3121** and the second upper protrusion **3122**, and may separate the upper surface of the second film **350** and the bottom surface of the first fire prevention sheet **360** apart from each other by the gap having the distance L**3**, thereby allowing gas generated from the battery cell C to be easily discharged. A chemical liquid may be sprayed from the first fire prevention sheet **360** toward the upper surface of the battery cell C where a fire occurred. Because the battery pack **10** includes the first spacer **370**, the first fire prevention sheet **360** and the second film **350** may be spaced apart from each other without forming a separate injection molding on the battery module **300** and the housing **100** or the cover **200** or without including an additional holder.

[0094] The first spacer **370** may not overlap other components of the second case **312**. In the embodiment illustrated in FIG. **6**, the first spacer **370** may not overlap with the first upper protrusion **3121**, the second upper protrusion **3122**, the insertion hole **3123**, the first fastening hole **3124**, or the second fastening hole **3125**. The first spacer **370** may not overlap the insertion hole **3123**, into which the battery cell C is inserted, such that the first spacer **370** is configured not to be damaged by gas and heat generated from the battery cell C. The first spacer **370** may be on the second film **350** and adjacent to an edge of the battery module **300** and spaced inward apart from the edge of the second film **350**. Therefore, the edges and center of the first fire prevention sheet **360** supported by the first spacer **370** may not sag downward.

[0095] The plurality of first spacers **370** may be included in each battery module **300**. The plurality of first spacers **370** may be along the edge of the case **310**. The plurality of first spacers **370** may be adjacent to the edge of the case **310** but may be spaced inwards from the edge of the case **310**. In one or more embodiments, the plurality of first spacers **370** may be adjacent to the long side portion and the short side portion of the second case **312**, but spaced apart from the long side and short side portion of the second case **312**. In the embodiment illustrated in FIG. **6**, one battery module **300** may include four first spacers **370**. However, the number of first spacers **370** is not limited to four, and the number of first spacers **370** may vary depending, for example, on the shape, size, weight, etc. of the first fire prevention sheet **360**. The first spacer **370** may include a flame-retardant material. In one or more embodiments, the first spacer **370** may be formed by stacking a plurality of flame-retardant materials. In one or more embodiments, the first spacer **370** may be formed by stacking a plurality of Poron™ tapes (e.g., a plurality of microcellular urethane or polyurethane sheets).

[0096] The second fire prevention sheet **380** may be in the lower portion of the battery module **300** and prevent (or at least mitigate) flames generated from the battery module **300** from spreading. In one or more embodiments, the second fire prevention sheet **380** may include a fire extinguishing agent, and in response to a flame occurring in one of the battery cells C, a portion of the second fire prevention sheet **380** may be configured to be torn or ruptured due to heat and pressure and the fire extinguishing agent may be sprayed into the battery cell C. The second fire prevention sheet **380** may include a flame-retardant material to prevent (or at least mitigate) flames from spreading to other components of the battery pack **10**, such as the substrate **400**. In one or more embodiments, the second fire prevention sheet **380** may include ceramic paper, mica, or aerogel. The second fire prevention sheet **380** may be below the first film **340** to cover the lower portion of the battery module **300**. The battery pack **10** may include one second fire prevention sheet **380**. In one or more embodiments, the second fire prevention sheet **380** may be on the bottom surface of the housing **100** to cover the entire bottom surface of the plurality of battery modules **300**. The second fire prevention sheet **380** may cover the three lower sides of the case **310**. In one or more embodiments, the second fire prevention sheet **380** may wrap the bottom surface of the case **310** (e.g., the bottom surface of the first case **311**) and two of the plurality of sides of the case **310**, which are connected to the bottom surface of the case **310** (e.g., two side surfaces corresponding to the long side portions of the first case **311**).

[0097] The second fire prevention sheet **380** may be spaced apart from the case **310** and/or the first film **340**. In the embodiment illustrated in FIG. **8**, the case protrusion **3111** of the first case **311** may extend into the housing protrusion **110** of the housing **100**, or the housing protrusion **110** may extend into the case protrusion **3111**, and the first film **340** and the second fire prevention sheet **380** may be spaced apart from each other by the distance L**4**. Thus, gas generated from the battery cell C may be easily discharged. By separating the second fire prevention sheet **380** and the first film **340** apart from each other, the fire extinguishing agent included in the second fire prevention sheet **380** may be easily sprayed over a relatively wide area.

[0098] The second fire prevention sheet **380** may include a second body portion **3801** and a second edge portion **3802**. The second body portion **3801** may be attached to the bottom surface of the first case **311** and may have a larger area than the case **310**. In one or more embodiments, the second body portion **3801** may cover the plurality of first cell tabs **331**. The second edge portion **3802** may extend upward from the edge or edges of the second body portion **3801**. In one or more embodiments, the second edge portion **3802** may correspond to long side portions of the second body portion **3801**. As illustrated in FIG. **3**, the second edge portion **3802** may extend upward from a pair of long side portions of the second body portion **3801** and surround at least a portion of the lower portion of the case **310**. The second edge portion **3802** may be configured to prevent (or at least mitigate) flames from spreading in the lateral direction (e.g., the Y-axis direction in FIG. **3**).

[0099] In the embodiment illustrated in FIG. **3**, the second edge portion **3802** may not be on the short side portions of the second body portion **3801**. The second edge portion **3802** may be on the pair of long side portions of the second body portion **3801** and may not be on the pair of short side portions of the second body portion **3801**. As illustrated in FIG. **3**, in the longitudinal direction of the battery module **300** (e.g., X-axis direction of FIG. **3**), the front surface and the rear surface of the battery module **300** or the case **310** may not be covered by the second fire prevention sheet **380**. The second fire prevention sheet **380** may cover three sides of the battery module **300** or the case **310** (e.g., the first case **311**). Short side portions of the case **310** may not be covered by the second fire prevention sheet **380**. Thus, the second fire prevention sheet **380** may prevent (or at least mitigate) the spread of flames generated from the long side portion of the battery module **300** and allow a small amount of gas to be easily discharged from the short side portions of the battery module **300**. Additionally, the flow of gas may be controlled such that the gas generated in the battery module **300** is discharged to the short side portions of the battery module **300**.

[0100] A lower end of the second edge portion **3802** may be below a center of the case **310** in a height direction and above a lower end of the case **310**. In one or more embodiments, the lower end of the second edge portion **3802** may be below the connecting portion between the first case **311** and the second case **312** and may be above the lower end of the first case **311**. If the second edge portion **3802** extends to the connection portion between the first case **311** and the second case **312**, the area covered by the second edge portion **3802** becomes too wide, causing an excessively high internal pressure of the battery module **300** if a flame occurs in the battery module **300**. In an embodiment in which the second edge portion **3802** is below the lower end of the first case **311**, the effect of preventing (or at least mitigating) the spread of a flame from the long side portion of the battery module **300** may decrease. An inner side surface of the second edge portion **3802** may be spaced apart from the side surface of the case **310**.

[0101] The second spacer **390** may connect or space apart a plurality of first fire prevention sheets **360** included in different battery modules **300** to/from each other. In the embodiment illustrated in FIG. **7**, the second spacer **390** may be on the long side portions (e.g., the first edge portion **3602**) of each first fire prevention sheet **360** included in different battery modules **300**, and may connect each first fire prevention sheet **360** to each other. The second spacer **390** may have both ends spaced inward relative to both ends of the first fire prevention sheet **360** in a longitudinal direction (e.g., an X-axis direction in FIG. **7**) (e.g., second spacer **390** may have a shorter length than the first fire prevention sheets **360** in the X-axis direction). The second spacer **390** may have a width L**1** (e.g., a length in the Y-axis direction of FIG. **7**) and a length L**2** (e.g., a length in the X-axis direction of FIG. **7**). The width L**1** may correspond to the separation distance between adjacent first fire prevention sheets **360**. The length L**2** may be less than a length of the first fire prevention sheet **360**. In one or more embodiments, the second spacer **390** may include a flame-retardant material. In one or more embodiments, the second spacer **390** may include the same material as that of the first film **340** and/or the second film **350**.

[0102] The substrate **400** may be electrically connected to the battery module **300** and may be configured to detect a current state of the battery module **300**, such as voltage and temperature, and to control the battery module **300**. In one or more embodiments, the substrate **400** may include a battery monitoring system (BMS). The substrate **400** may be connected to a plurality of battery cells C through the bus bar **320**. The substrate **400** may be configured to detect a voltage or temperature of the plurality of battery cells C and to control the battery module **300** to prevent over-voltage, over-current, and/or over-discharge of the battery module **300**. The substrate **400** may include a plurality of circuits and components. The substrate **400** may be a printed circuit board (PCB).

[0103] The substrate **400** may be on the battery module **300**. In one or more embodiments, as illustrated in FIGS. **3** and **4**, the substrate **400** may be on the first fire prevention sheet **360**. The substrate **400** may be spaced apart upward in a height direction (e.g., the Z-axis direction in FIG. **3**) from the first fire prevention sheet **360** without contacting the first fire prevention sheet **360**. Therefore, heat generated in the battery module **300** may be prevented (or at least mitigated) from being directly transferred to the substrate **400**. The substrate **400** may be supported on the case **310**. In one or more embodiments, the substrate **400** may be supported on the first upper protrusion **3121** and the second upper protrusion **3122** of the second case **312**. The first upper protrusion **3121** and

the second upper protrusion **3122** may separate the substrate **400** apart upward from the first fire prevention sheet **360** and extend into the substrate **400**.
[0104] Although the present disclosure has been described with reference to the embodiments illustrated in the drawings, these are merely examples. Those skilled in the art may fully understand that various modifications and equivalent other embodiments may be made from the embodiments. Therefore, the true technical protection scope of the present disclosure should be determined based on the attached claims.
[0105] According to the battery pack according to the embodiments, a fire may be quickly extinguished and the spread of fire may be prevented (or at least mitigated) without forming an additional injection molding or applying additional components to other components of the battery pack. According to the battery pack according to the embodiments, sagging of a sheet for fire extinguishing and fire prevention of a battery module may be prevented (or at least mitigated).
[0106] However, the effects that may be achieved through the present disclosure are not limited to the above-described effects, and other technical effects not mentioned herein may be clearly understood by those skilled in the art from the description of the present disclosure described below.
[0107] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims.

Claims

1. A battery pack comprising: a housing; and a plurality of battery modules in the housing, wherein each of the plurality of battery modules comprises: a plurality of battery cells; a case accommodating the plurality of battery cells; a first fire prevention sheet on the case; and a first spacer on the case and configured to separate the first fire prevention sheet from an upper surface of the case.
2. The battery pack as claimed in claim 1, wherein the first fire prevention sheet comprises a plurality of first fire prevention sheets, wherein the plurality of battery modules further comprises a second spacer configured to connect the plurality of first fire prevention sheets to each other, and wherein the plurality of first fire prevention sheets is spaced apart from each other.
3. The battery pack as claimed in claim 2, wherein the plurality of first fire prevention sheets correspond to the plurality of battery modules.
4. The battery pack as claimed in claim 2, wherein the second spacer comprises a flame-retardant material, and wherein both ends of the second spacer in a longitudinal direction are spaced inward relative to both ends of the first fire prevention sheet.
5. The battery pack as claimed in claim 1, wherein the first spacer comprises a plurality of first spacers, and wherein the plurality of first spacers is along an edge of the case.
6. The battery pack as claimed in claim 5, wherein the plurality of first spacers is adjacent to an edge of the case and are spaced apart inwards from the edge.
7. The battery pack as claimed in claim 1, wherein the first spacer comprises a flame-retardant material comprising a plurality of stacked layers, and wherein an upper surface of the first spacer is in contact with a bottom surface of the first fire prevention sheet.
8. The battery pack as claimed in claim 1, wherein the case comprises: a first case; and a second case connected to the first case, the first case and the second case accommodating the plurality of battery cells, wherein the second case comprises a protrusion on an upper surface of the second

case, wherein the protrusion supports the first fire prevention sheet, and wherein the first spacer does not overlap the protrusion.

9. The battery pack as claimed in claim 8, further comprising a substrate on the plurality of battery modules and electrically connected to the plurality of battery modules, and wherein the protrusion separates the substrate from the upper surface of the second case.

10. The battery pack as claimed in claim 9, wherein the protrusion comprises: a plurality of first upper protrusions on the upper surface of the second case; and a plurality of second upper protrusions on the upper surface of the second case and spaced apart from the plurality of first upper protrusions, wherein at least some of the plurality of first upper protrusions and at least some of the plurality of second upper protrusions are inserted into the substrate and configured to support the substrate to separate the substrate from the upper surface of the first fire prevention sheet.

11. The battery pack as claimed in claim 8, wherein each of the plurality of battery modules comprises: a first film covering a bottom surface of the first case; and a second film covering the upper surface of the second case, wherein a plurality of first spacers is on the first film.

12. The battery pack as claimed in claim 11, wherein the second film is spaced apart from a bottom surface of the housing.

13. The battery pack as claimed in claim 1, wherein the first fire prevention sheet comprises a flame-retardant material or a fire extinguishing agent, and wherein the first fire prevention sheet extends around three sides of an upper portion of the case.

14. The battery pack as claimed in claim 13, wherein the first fire prevention sheet comprises: a first body portion covering the upper surface of the case; and two first edge portions extending downward from two long side portions of the first body portion, wherein short side portions of the case are not covered by the first fire prevention sheet.

15. The battery pack as claimed in claim 14, wherein a lower end of each of the two first edge portions is above a center of the case in a height direction and below an upper end of the case.

16. The battery pack as claimed in claim 1, wherein the case comprises a case protrusion protruding downward from a bottom surface of the case, and wherein the housing has a bottom surface spaced apart from the bottom surface of the case by the case protrusion.

17. The battery pack as claimed in claim 16, wherein the housing comprises housing protrusions on the bottom surface of the housing, and wherein the housing protrusion is in the case protrusion or the case protrusion is in the housing protrusion.

18. The battery pack as claimed in claim 1, wherein each of the plurality of battery modules further comprises a second fire prevention sheet on a bottom surface of the housing and covering all bottom surfaces of the plurality of battery modules.

19. The battery pack as claimed in claim 18, wherein the second fire prevention sheet comprises a flame-retardant material or a fire extinguishing agent, and wherein the second fire prevention sheet extends around three sides of a lower portion of the case.

20. The battery pack as claimed in claim 19, wherein the second fire prevention sheet comprises: a second body portion covering a bottom surface of the case; and two second edge portions extending upward from two long side portions of the second body portion, wherein short side portions of the case are not covered by the first fire prevention sheet.
